

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Eighth Semester of B. Tech (EE) Examination****April-May 2018****EE407 Commissioning of Electrical Equipments****Date: 02.05.2018, Wednesday****Time: 01.30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I**Q - 1 Answer the questions below. [14]**

- a. Give the classification of Routine test, type test and special test of transformers and explain any two in detail. [07]
- b. Explain testing methodology of mineral oil of transformer. [07]

OR

- a. Give the classification of Routine test, type test and special test of Induction Motors and explain any two in detail. [07]
- b. Explain the troubleshooting and maintenance of dc machine. [07]

Q – 2 Answer the questions below. [14]

- a. Explain OC, SC and Earth Fault test of armature and field coils of dc machine. [07]
- b. Explain Hopkinson test or back to back test of dc machine. [07]

OR

- a. Explain Moisture proofness, leakage current and pull out torque test of induction motor. [07]
- b. Explain the troubleshooting and maintenance of induction motor. [07]

Q - 3 Answer any one. [07]

- a. Explain open circuit and short circuit test of a three phase synchronous generator. [07]
- b. Explain various methods to measure insulation resistance. [07]
- c. Explain different types of Isolators. [07]

SECTION – II**Q – 4 Answer the questions below. [07]**

- a. What is the typical frequency of high frequency electronic ballast? [01]

(a) 50 Hz	(b) 50 kHz
(c) 30 kHz	(d) 60 Hz

- b. For large low voltage transformers, the most commonly used connections are [01]
 (a) Star - star connection (b) Delta - delta connection
 (c) V connection (d) All of these
- c. Which equipment is used for EHV lines to improve power transferability? [01]
 (a) Shunt capacitor (b) Shunt reactor
 (c) Series capacitor (d) All of these
- d. Which of these can improve load factor of an electrical system? [01]
 (a) Increased average load (b) Decreased average load
 (c) Increased maximum load (d) Increased maximum demand
- e. Perfect conductor has [01]
 (a) zero conductivity (b) unity conductivity
 (c) infinite conductivity (d) none of the above
- f. What is ingress protection (IP Protection) or Degree of protection in DC machine? [02]
- Q – 5.a** Explain short time current rating test on circuit breakers. [04]
- Q – 5.b** Answer any two questions. [10]
- (i) Explain type tests on Lightning arrestors. [05]
 (ii) What is thermography? State its advantage and disadvantages. [05]
 (iii) Explain commissioning steps of substation equipments. [05]
- Q - 6** Answer any two. [14]
- a. Explain briefly Step potential, Touch potential and Transfer potential. What is earthing grid? [07]
 b. Explain Flick test, Magnetization curve test and Ratio test on CT. [07]
 c. Following are the details of induction generator and as a squirrel cage motor: [07]
 Supply 440 volts, No of poles = 4, kW rating 30kW, Full load speed 1760 rpm, rated current 41A, full load power factor 0.85.
 Calculate capacitance of each delta connected capacitor, speed of petrol engine having frequency of 60 Hz.
